

Saurav Kumar

+916206846984 | sauravx25@gmail.com | [linkedin](#) | [github](#) | [portfolio](#)

Education

National Institute of Technology, Warangal

Nov 2021 - May 2025

Bachelor of Technology in Electronics and Communication Engineering

Experience

SAP Labs India

July 2025 - Present

Associate Developer

Bengaluru, India

- Built **CAP backend with OData services** and **Spring Boot APIs**, orchestrating **600+ automated tests** across **5 SAP HANA clusters** with dynamic dependency handling
- Architected a **multi-target, API-first cloud framework** on **SAP BTP** with locking and DB isolation across **200+ concurrent runs**, cutting failures **30%** and cycle time **50%**
- Designed an **agentic LLM Gateway** with **mode-aware agents**, **tool-use orchestration** over 50+ CAP tools, **prompt caching**, and a session-scoped fact store also powering async **failure classification at 93.5% accuracy avg** and **auto-healing**, reducing manual triage by **60%**
- Built a **streaming SSE chat pipeline** on the gateway with conversation history and shareable chat links, enabling an in-app assistant for iterative deep-dives over execution data
- Developed a **React + UI5 Web Components UI** with Dashboards and Custom Reports for test analysis, centralized configuration system, and branch-based execution

SAP Labs India

May 2024 - July 2024

SDE Intern

Bengaluru, India

- Built a deepfake detection pipeline (EfficientNet-B4) trained on **140k+ samples**, achieving **98% validation accuracy**
- Built a Flask inference service supporting **300+ concurrent requests** with real-time image/video processing; optimized 1080p inference to **22 FPS**
- Tech: **Python, TensorFlow, Flask, OpenCV, WebRTC, React**

Projects

Railway Ticket Booking System

[GitHub](#)

- Designed a **Spring Boot application** with PostgreSQL, Redis, Apache Kafka, and Elasticsearch to handle high-concurrency train bookings
- Built a **distributed seat allocation system** using Redis Lua scripts and TTL-based locks, combined with idempotent APIs and optimistic concurrency control to prevent race conditions and overbooking
- Designed an **event-driven workflow** using Kafka (choreography) for cancellations, refunds, and waitlist promotion with retry handling and dead-letter queues
- Developed a **CQRS-based search pipeline** (PostgreSQL to Elasticsearch sync) with Redis caching, achieving sub-second train search latency
- Secured services using **JWT-based RBAC**, exposed 30+ REST APIs via Swagger, and built a frontend using React and shadcn/ui for booking and search flows

Technical Skills

Languages: **Java, C++, JavaScript, Python, React, SQL**

Backend & Frameworks: **Spring Boot, Node.js, SAP CAP, Express, Elasticsearch, LangGraph**

Databases & Messaging: **PostgreSQL, MySQL, Redis, Kafka**

Cloud & DevOps: **Docker, AWS (EC2, S3, SQS), Jenkins, Git**

Core Concepts: **Operating system, OOP, DSA, Networking, Concurrency, Design**

Achievements

- **99.32%ile** among **10+ lakh** candidates in **JEE Mains (Mar 2021)**.
- **NTSE Qualified** and **District Topper** in Class X(97.6%) also secured 95.8% in Class XII (CBSE)
- Solved **500+ algorithmic** problems on LeetCode, Codeforces, and InterviewBit.